Sockets

Aim:

To implement echo client server using TCP/IP sockets

procedure:

// To write the Tcp code

Import Socket

Import threading

Server

def Start_Server():

Server = Socket.Socket (Socket.AF_INET, Socket.SOCK_STREAM)

Server.bind (('127.0.0.1', 5000))

Server.listen()

print ("[Server] Listening on 127.0.0.1:5000")

Conn, addr = Server.accept()

print(f"[Server] Connected {addr}")

while True:

data = Conn.recv(1024).decode()

if not data or data.lower() == 'quit':

print("[Server] Connect closing the line

break

```
print(f"[SERVER] Received: {data}")
conn.send(f"Server received {data}".encode())
Connection.close()
Server.close()

# client ...
def start_client():
    client = Server.socket(socket.AF_INET, socket.SOCK_
    STREAM)
    client.connect(('127.0.0.1', 5000))
    print("[client] Connected to Server").
    while True:
        msg = input("Enter message (or 'quit to exit):"
        client.close()
# ... Main ....
    if __name__ == "__main__";
    Server_thread = thread.Thread(target=start_Server,
    daemon = True)
    Server_thread.start()
#start_client().
```

Result:

Therefore, the implement echo client server using TCP/UDP socket had successfully executed.

~~12(b).~~

output

Server listening on 127.0.0.1:5000 (client) Connected to Server.

[Server] Connected by ('127.0.0.1', 53458)

HELLO

Hi.

[Server] connection closed by client

Result:

Therefore, the implement echo client / server TCP/UDP sockets had executed successfully.

12b b) Implement a chat program using Sockets programming

Aim:

To Implement a chat program using code Sockets programm

procedure

1) TO Implement the UDP code.

```
Import socket
Import threading
# ... Server ...
def udp_Server():
    S = Socket.Socket(Socket.AF_INET, SOCKET SOCK_DR
    S.bind(('127.0.0.1', 12000))
    print("[Server] Running on 127.0.0.1:12000")
    while True:
        data, addr = S.recvfrom(1024)
        msg = data.decode()
        if msg.lower() == 'quit':
print(f"[Server] client {addr} ended the
Connect.").
            break
print(f"[Server] Recieved from {addr}: {msg}
S.sendto(data, addr)
S.close
print("[Server] Stopped.")
```

```
# client:
def udp_client():
    c = socket.socket(socket.AF_INET, SOCK_DGRAM)
    server = ('127.0.0.1', 12000)
    while True:
        msg = input("Enter message (or 'quit' to stop): ")
        c.sendto(msg.encode(), server)
        if msg.lower() == 'quit':
            print("[client] closing connection...")
            break
        data, _ = c.recvfrom(1024)
        print("Echo from server:", data.decode())
    c.close()
    print("[client] stopped.")
# --- Main ---
if __name__ == "__main__":
    threading.Thread(target=udp_server,
        daemon=True).start()
    udp_client()
```

Output:

```
Enter message (or 'quit' to stop): [server]
Running on 127.0.0.1:12000
hi
[server] Received from ('127.0.0.1', 49670): hi
```

Enter from server: hi

Enter message (or 'quit' to stop): ('127.0.0.1', 49670):

[client] closing connection -- [Server] client ('127.0.0.1', 49670) ended the connection

[client] Stopped [server] Stopped.

Result:

Therefore, to Implement a chat program using Socket programming executed Successfully.